STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



AI

DECISIONS WITH DATA

How far two experts disagree on the same case, what deciding with data is worth, when a formula beats the expert, why people drop an algorithm, and a card for a decision with data.

55% was the median gap between the premiums two underwriters set for the same case; the executives had expected about 10%.

KAHNEMAN ET AL., 2021

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NO. 20 · AI

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Managers get more data every year than they can read. This issue asks where data actually improves a decision: where it replaces noise, where it informs judgment, and where people stop trusting it.

In one insurer, the median gap between two underwriters pricing the same case was 55%. Among 179 large firms, those that decided with data had output and productivity 5–6% higher than their other investments and IT would predict. In studies of prediction, formulas were more accurate than experts on average, yet people abandon an algorithm once they see it err, unless they may adjust it a little.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

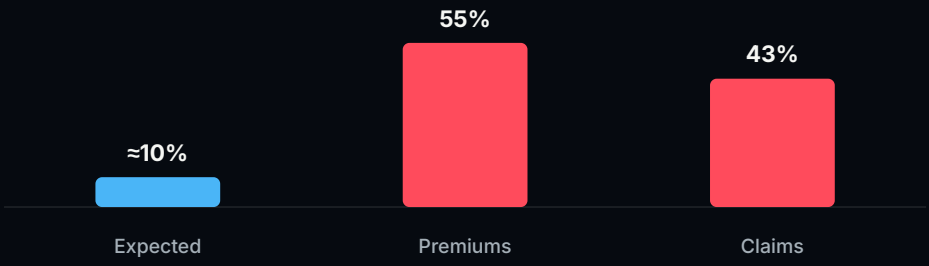
Practice
The steps and the card are proposals to try, not research results.

COVER STORY

The noise audit

In a noise audit at an insurance company, described by Daniel Kahneman, Olivier Sibony and Cass Sunstein, underwriters priced the same cases, each on their own, and claims adjusters valued the same claims.

MEDIAN GAP BETWEEN TWO PROFESSIONALS JUDGING THE SAME CASE



Kahneman and his co-authors call the method a noise audit: people in the same unit judge the same cases, and how far their answers differ is the measure of noise. Their remedies are algorithms, or procedures that push judgments towards consistency.

OUR READING

Noise is invisible until two people judge the same case. Most teams never do that, so they never see it.

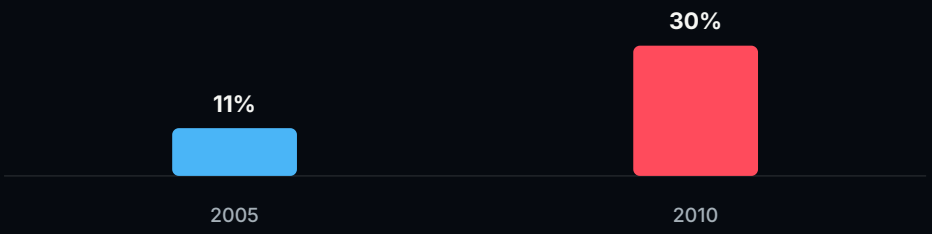
Sources: Daniel Kahneman, Olivier Sibony & Cass R. Sunstein, 2021 (via strategy+business, 2021; book passages quoted by readers); Daniel Kahneman et al., Harvard Business Review, 2016

THE NUMBERS

What deciding with data is worth

Erik Brynjolfsson, Lorin Hitt and Heekyung Kim studied how 179 large listed firms make decisions. Those that decided with data had output and productivity 5–6% higher than their other investments and use of IT would predict.

US MANUFACTURING PLANTS THAT DECIDE WITH DATA



In US manufacturing, Brynjolfsson and Kristina McElheran found that the share of plants deciding with data almost tripled in five years. Among the data leaders of large firms surveyed by Wavestone, 48.1% called their organisation data-driven in 2024, up from 23.9% a year earlier.

OUR READING

Data pays when it changes how decisions are made, not when it fills more reports.

Sources: Erik Brynjolfsson, Lorin M. Hitt & Heekyung Hellen Kim, SSRN, 2011; Erik Brynjolfsson & Kristina McElheran, American Economic Review, 2016 (via Abstract at AEA); Wavestone, 2024

The firm study is observational, not an experiment. The second study counts plants, not firms. Wavestone's figures are data leaders' own view, not a representative sample.

THE MODEL

Rule, judgment, or both

Data can enter a decision in three ways. Which one fits depends on how often the decision repeats and how differently people answer the same case.

01	**A rule decides**	For frequent, similar cases with a clear outcome: the formula gives the answer, people handle the exceptions.
02	**A rule proposes, a person adjusts**	The formula gives a figure; the person may move it within a set range and writes down why.
03	**A person decides, with data in front**	For rare or new decisions: the data is evidence, the judgment is the person's, and it is written down so it can be checked later.

HYPOTHETICAL EXAMPLE, A WEEKLY STAFFING CALL

Rule	Staff = forecast orders ÷ 120 per person
Adjust	The shift lead may add or remove up to 2 people
Record	Each change, with one line on why

After a month, the record shows whether the changes helped. The figures are invented.

OUR READING

Data does not replace judgment. It replaces the noise in it: the part that depends on who decides, and on which day.

Source: Berkeley J. Dietvorst, Joseph P. Simmons & Cade Massey, Management Science, 2018

The three ways are the editors' summary; the second follows Dietvorst, Simmons and Massey (2018).

More in the essay KPIs do not improve in Excel

WHAT THE RESEARCH SAYS

The formula and the expert

William Grove and colleagues compared the judgment of experts with mechanical, statistical prediction across studies of prediction in health and human behaviour.

≈10%
more accurate on average: the formula

33–47%
of studies: the formula clearly better

6–16%
of studies: the expert clearly better

Yet people are reluctant to use formulas. In five studies, Berkeley Dietvorst, Joseph Simmons and Cade Massey found that people who saw an algorithm err trusted it less and chose it less often than a worse human forecaster, even when it had done better. They lost faith in it faster than in a person after the same mistake.

When people could change the algorithm's forecast a little, they used it much more often and did better, even when the change allowed was very small.

OUR READING

Let people adjust the formula a little and write down why. The formula stays in use, and the adjustments stay visible.

Sources: William M. Grove et al., Psychological Assessment, 2000 (via Abstract at PubMed); Berkeley J. Dietvorst, Joseph P. Simmons & Cade Massey, Journal of Experimental Psychology: General, 2015; Berkeley J. Dietvorst, Joseph P. Simmons & Cade Massey, Management Science, 2018

In many studies the two were about as accurate. The ranges depend on how a clear difference is counted.

HOW IT IS MEASURED

A noise audit in your team

The audit can be run on a small scale: the same cases, judged separately by the people who usually decide them.

1. **Choose a repeated decision**
   One that several people make: a priority, an estimate, a grade.
2. **Prepare five real cases**
   Written up the same way, without names.
3. **Let each person judge alone**
   No discussion until all the answers are in.
4. **Look at the spread**
   For each case, the gap between the highest and the lowest answer.
5. **Agree a rule where it is widest**
   Write down how such a case is decided, and test again in three months.

HYPOTHETICAL EXAMPLE, FIVE SHIFT LEADS ESTIMATE THE SAME DELAYS

Case 1	from 20 to 45 minutes
Case 2	from 10 to 15 minutes
Case 3	from 30 to 90 minutes

Case 3 needs a rule first. The cases and numbers are invented.

Source: Daniel Kahneman et al., Harvard Business Review, 2016

The steps and the example are the editors', after the noise audit described by Kahneman and colleagues.

TOOL OF THE ISSUE

The data decision card

One card for a decision that repeats. Fill it in with the people who make it, and look at it again after a month.

01 The decision what is decided, how often, by whom

02 The data which figures, from where, how fresh

03 The rule how the data becomes an answer

04 What a person may change by how much, with the reason written down

05 Noise test date, five cases, who judges them

06 Review after a month: did the changes help?

Sources: Daniel Kahneman et al., Harvard Business Review, 2016; Berkeley J. Dietvorst, Joseph P. Simmons & Cade Massey, Management Science, 2018

A practice proposed by the editors, after the noise audit of Kahneman and colleagues and the adjustable algorithms of Dietvorst, Simmons and Massey.

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Noise: A Flaw in Human Judgment
Daniel Kahneman, Olivier Sibony & Cass R. Sunstein, 2021 · via strategy+business, 2021; book passages quoted by readers

Noise: How to Overcome the High, Hidden Cost of Inconsistent Decision Making
Daniel Kahneman et al., Harvard Business Review, 2016
https://hbr.org/2016/10/noise

Strength in Numbers: How Does Data-Driven Decisionmaking Affect Firm Performance?
Erik Brynjolfsson, Lorin M. Hitt & Heekyung Hellen Kim, SSRN, 2011
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1819486

The Rapid Adoption of Data-Driven Decision-Making
Erik Brynjolfsson & Kristina McElheran, American Economic Review, 2016 · via Abstract at AEA
https://doi.org/10.1257/aer.p20161016

2024 Data and AI Leadership Executive Survey
Wavestone, 2024
https://www.wavestone.com/en/insight/data-ai-executive-leadership-survey-2024/

Clinical versus mechanical prediction: A meta-analysis
William M. Grove et al., Psychological Assessment, 2000 · via Abstract at PubMed
https://doi.org/10.1037/1040-3590.12.1.19

Algorithm aversion: People erroneously avoid algorithms after seeing them err
Berkeley J. Dietvorst, Joseph P. Simmons & Cade Massey, Journal of Experimental Psychology: General, 2015
https://doi.org/10.1037/xge0000033

Overcoming Algorithm Aversion: People Will Use Imperfect Algorithms If They Can (Even Slightly) Modify Them
Berkeley J. Dietvorst, Joseph P. Simmons & Cade Massey, Management Science, 2018
https://doi.org/10.1287/mnsc.2016.2643

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

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